

Fine-tuning leaves path-dependent structure in model weights. We estimate the resulting model phylogeny, score it against known training trees, and test whether weight proximity predicts behavioral similarity. We evaluate four-point additivity as a label-free diagnostic of clade recovery.

### A. Controlled recovery (known truth)

true tree  $\mathcal{P}^*$

recovered  $\hat{\mathcal{P}}$

weights  $\rightarrow$  NJ

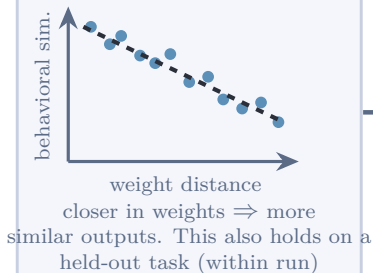
RF / FN



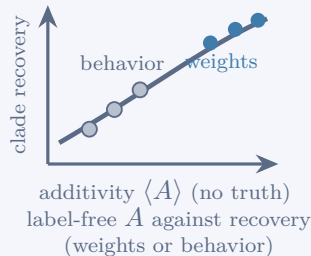
✓ better than random tree

neighbor-joining estimate,  
scored against known truth

### B. Weights predict behavior random-effects pooled $r_{DL} < 0$



### C. Additivity and recovery



full weight access,  
known lineage

weight  $\rightarrow$  behavior link

diagnostic without known truth